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1.

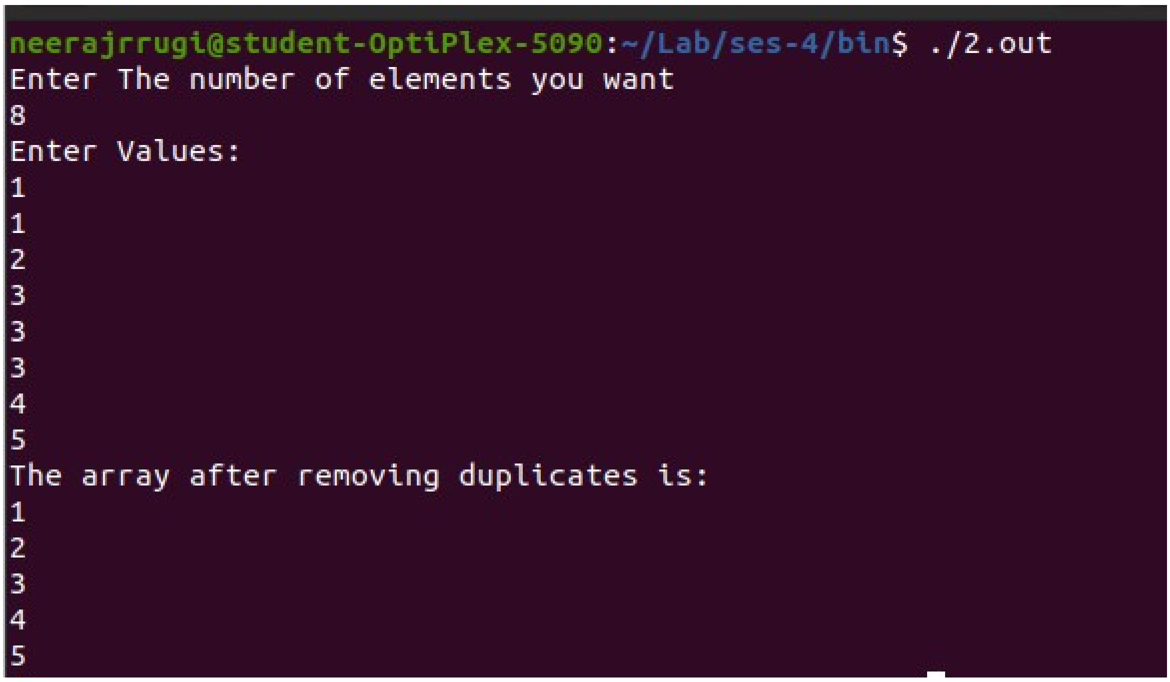
```
#include<stdio.h>
int main(){
	printf("Enter The number of elements you want\n");
	int len = 0;
	scanf("%d", &len);
	int a[len];
	int out[len][2];
	int temp;
	int alraedy_present = 0;
	int ele = 0;

	printf("Enter Values: \n");
	for(int i = 0; i < len; i++){
		scanf("%d", &a[i]);
	}
	for(int i = 0; i < len; i++){
		temp = a[i];
		for(int j= 0; j < len;j++){
			if(temp == out[j][0]){
				alraedy_present = 1;
				out[j][1] += 1;
				break;
			}
		}
		if(!alraedy_present){
				out[ele][0] = temp;
				out[ele][1] = 1;
				ele++;
			}
		alraedy_present = 0;
	}
	for(int i = 0; i < ele; i++){

		printf("%d Occours: %d times\n", out[i][0], out[i][1]);
	}
	return 0;
}
```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./1.out
Enter The number of elements you want
8
Enter Values:
1
1
23
34
34
34
5
6
1 Occours: 2 times
23 Occours: 1 times
34 Occours: 3 times
5 Occours: 1 times
6 Occours: 1 times
```

2.

```
#include<stdio.h>
int main(){
	printf("Enter The number of elements you want\n");
	int len;
	scanf("%d", &len);
	int a[len];
	int out[len];
	int temp;
	int alraedy_present = 0;
	int ele = 0;

	printf("Enter Values: \n");
	for(int i = 0; i < len; i++){
```

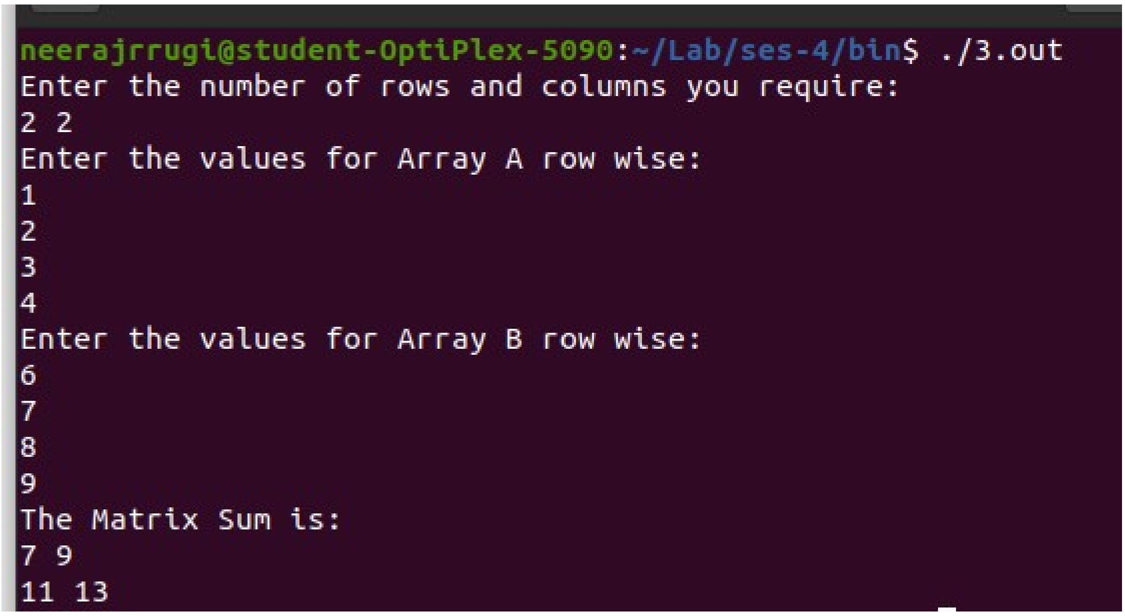
```
        scanf("%d", &a[i]);
    }
    for(int i = 0; i < len; i++){
        temp = a[i];
        for(int j= 0; j < len;j++){
            if(temp == out[j]){
                alraedy_present = 1;
                break;
            }
        }
        if(!alraedy_present){
                out[ele] = temp;
                ele++;
            }
        alraedy_present = 0;
    }
    printf("The array after removing duplicates is: \n");
    for(int i = 0; i < ele; i++){

        printf("%d\n", out[i]);

    }
    return 0;
}
```



```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./2.out
Enter The number of elements you want
8
Enter Values:
1
1
2
3
3
3
4
5
The array after removing duplicates is:
1
2
3
4
5
```

3.

```
#include<stdio.h>
int main(){
	printf("Enter the number of rows and columns you require: \n");
	int r,c;
	r = 0;
	c = 0;
	scanf("%d %d", &r, &c);
	printf("Enter the values for Array A row wise: \n");
	int a[r][c], b[r][c], x[r][c];
	for(int i = 0; i < r; i++){
		for(int j = 0; j < c; j++){
			scanf("%d", &a[i][j]);
		}
	}
	printf("Enter the values for Array B row wise: \n");
	for(int i = 0; i < r; i++){
		for(int j = 0; j < c; j++){
			scanf("%d", &b[i][j]);
		}
	}
	for(int i = 0; i < r; i++){
		for(int j = 0; j < c; j++){
			x[i][j] = a[i][j] + b[i][j];
		}
	}
	printf("The Matrix Sum is: \n");
	for(int i = 0; i < r; i++){
		for(int j = 0; j < c; j++){
			printf("%d ", x[i][j]);
		}
		printf("\n");
	}
}
```



```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./3.out
Enter the number of rows and columns you require:
2 2
Enter the values for Array A row wise:
1
2
3
4
Enter the values for Array B row wise:
6
7
8
9
The Matrix Sum is:
7 9
11 13
```

4.

```
#include<stdio.h>
int main(){
	printf("Enter the order of the matrix \n");
	int r;
	scanf("%d", &r);
	printf("Enter the values for Array A row wise: \n");
	int a[r][r];
	int p_dig =0, s_dig = 0;

	int x = 0, y = r-1;
	for(int i = 0; i < r; i++){
		for(int j = 0; j < r; j++){
			scanf("%d", &a[i][j]);
		}
	}
	for(int i = 0; i < r; i++){
		p_dig += a[i][i];
	}


	for(int i = 0; i < r; i++){
		s_dig += a[x][y];
		x++;
		y--;
	}

	printf("The Primary diagonal Sum is: %d\n",p_dig);
	printf("The Secondary diagonal Sum is: %d\n",s_dig);
}
```

```
neerajrrugi@student-OptiPlex-5090:~/Lab/ses-4/bin$ ./4.out
Enter the order of the matrix
2
Enter the values for Array A row wise:
2
1
3
5
The Primary diagonal Sum is: 7
The Secondary diagonal Sum is: 4
```